University Examination Series: Python Programming (PYQ)

Combined Past Examination Papers (2022 - 2025) | Total Marks: 100 | Time: 3 Hours

Annual Examination 2024 — Question Paper

Section A: Conceptual & Analytical Short Answer Questions (Answer Any 5 — 5 Marks Each)

1. Differentiate between deep copy and shallow copy with memory schematics. In what practical scenarios is deepcopy required? (5 marks)

2. Explain the LEGB rule governing variable resolution in Python. What is the precise purpose of the nonlocal keyword? (5 marks)

3. Explain why mutable default arguments in function definitions produce unexpected side-effects. How do you resolve this pattern? (5 marks)

4. Differentiate between a generator function and a standard function. What are the memory advantages of yield over return? (5 marks)

5. Explain the Global Interpreter Lock (GIL) in CPython. Why does multithreading fail to accelerate CPU-bound computations? (5 marks)

6. Write short notes on dictionary comprehension and set comprehension with practical code snippets. (5 marks)

Section B: Long-Form Comprehensive Problems & Architecture (10 & 15 Marks Each)

7. Explain Object-Oriented Programming principles in Python with detailed code implementations: a) How does Python achieve encapsulation without explicit private keywords? Explain name mangling. (5 marks) b) Describe the C3 Linearization Algorithm and Method Resolution Order (MRO) in multiple inheritance. (5 marks) c) Implement a custom class that overloads the +, ==, and len operators using special dunder methods. (5 marks)

8. Detailed study of Exception Handling and Context Managers: a) Explain the complete lifecycle of try-except-else-finally blocks. When does the else block execute? (5 marks) b) Create a custom exception class `InsufficientFundsError` that records account balance and withdrawal attempt. (5 marks) c) Implement a custom context manager class `DatabaseConnectionManager` utilizing `__enter__` and `__exit__` to ensure atomic commits and automatic rollbacks upon exception. (5 marks)

9. Advanced Functional Programming and Decorators: a) Explain the concept of First-Class Functions and Closures in Python. (5 marks) b) Write a parameterized decorator `@retry(max_attempts=3, delay=2)` that intercepts network exceptions and retries the decorated function before raising failure. (10 marks)

10. File Processing and Systems Programming: a) Write a Python program to read a 10 GB log file line-by-line using a generator without loading the entire file into RAM, filtering for HTTP 500 error codes, and dumping IP frequency counts into a JSON report. (10 marks)